

Seminar on EU's Instantaneous Events
First Reading-Report



Department of Computer Science and
Information Technology
Min-Sheng, Tsai-蔡旻盛

Table of Contents

Hua Wei data privacy:	2
Global_Wafers (TSMC) Move to Germany.....	4
Reference	6

Hua Wei data privacy:

Huawei Cloud is making good inroads into the South African market thanks to its [security](#), in-country data centers, and support.

Reflection on [Huawei Data Privacy Concerns](#)

In recent years, Huawei has been at the center of numerous controversies, with one of the most significant being concerns over data privacy. As the Chinese tech giant has expanded its global reach, questions about how it handles user data and its relationship with the Chinese government have become increasingly prevalent. Reflecting on this issue prompts considerations not only about Huawei's practices but also about broader implications for data privacy and international relations.

At the heart of the Huawei data privacy debate [lies the concern that the company may be subject to Chinese government influence or control](#), raising questions about the security and privacy of user data. [The fear is that Huawei could be compelled to provide sensitive information to Chinese authorities](#), either voluntarily or through legal mandates, posing risks to user privacy and national security in countries where Huawei operates.

[This issue raises fundamental questions](#) about the balance [between national security interests](#) and the [principles of data privacy](#) and [user rights](#). Governments around the world are grappling with how to address these concerns while also benefiting from the technological advancements and economic opportunities that companies like Huawei offer. It underscores the need for robust regulatory frameworks and international cooperation to safeguard data privacy in an increasingly

interconnected world.

From a personal perspective, the Huawei data privacy concerns serve as a reminder of the importance of being vigilant about the technologies and platforms we entrust with our data. As individuals, [we must be aware of the potential risks and take steps to protect our privacy](#), whether through understanding privacy policies, using encryption tools, or advocating for stronger data protection laws.

In conclusion, [the Huawei data privacy concerns](#) represent a [multifaceted issue with far-reaching implications](#) for individuals, governments, and the global tech industry. Addressing these concerns requires a balanced approach that upholds both privacy rights and national security interests while fostering transparency, accountability, and cooperation among stakeholders. As we navigate the complexities of the digital age, it is imperative that we remain alert and engaged in shaping policies and practices that promote a safe and secure digital environment for all.

Global_Wafers (TSMC) Move to Germany

It was Feb. 1, 2022, the first day of the Year of the Tiger. The clock was winding down, and when it hit 6:59 a.m. in Taiwan, Global_Wafers Co. Chairperson Doris Hsu (徐秀蘭) and the many other executives gathered were extremely tense but ready for whatever was about to happen.

The recent decision by Global Wafers, a major player in the semiconductor industry and parent company of Taiwan Semiconductor Manufacturing Company (TSMC), to establish a manufacturing facility in Germany marks a significant development in the global tech landscape. **This move prompts considerations about the factors driving such a decision and the potential implications** for both the **semiconductor industry** and the **broader geopolitical context**.

At the forefront of this decision is the growing recognition of the importance of semiconductor manufacturing capabilities in ensuring national and economic security. **As the world becomes increasingly reliant on semiconductor chips for a wide range of applications**, including **smartphones, automobiles**, and **critical infrastructure**, countries are seeking to bolster their domestic semiconductor industries to reduce dependence on foreign suppliers and mitigate supply chain risks.

Germany's appeal as a destination for Global Wafers' new manufacturing facility reflects the country's strengths in engineering, innovation, and skilled workforce. By establishing a presence in Germany, **Global Wafers aims to leverage these strengths to expand its production capacity and meet the growing demand for semiconductor chips in Europe and beyond**.

From a broader Global_Wafer's perspective, the Global_Wafer's move to Germany underscores the shifting dynamics of global tech competition and the rise of new players in the semiconductor industry. Traditionally dominated by

companies in East Asia, particularly Taiwan, South Korea, and China, the semiconductor market is witnessing the emergence of new manufacturing hubs and players in Europe and North America.

From a personal standpoint, the Global_Wafers move to Germany prompts reflection on the interconnectedness of global economies and the role of technology in driving innovation and economic growth. As consumers and citizens, we rely on semiconductor chips in virtually every aspect of our daily lives, from communication and transportation to healthcare and entertainment. Understanding the factors influencing semiconductor manufacturing decisions helps us appreciate the complexities of the global tech ecosystem and the implications for future technological advancements and societal progress.

In conclusion, Global_Wafers' decision to establish a manufacturing facility in Germany highlights the evolving nature of the semiconductor industry and the strategic importance of semiconductor manufacturing capabilities in today's interconnected world. As countries and companies navigate the challenges and opportunities of the digital age, collaboration, innovation, and strategic investments will be essential in shaping the future of technology and ensuring a more resilient and prosperous global economy!

Reference

1. <https://www.wealth.com.tw/articles/0aa4acb1-89ba-4b56-9d36-9aac2cb6a4f5>
2. <https://www.itweb.co.za/article/data-security-and-sovereignty-huawei-cloud-goes-to-great-lengths-to-protect-sa-data/WnpNgq21oolMVrGd>
3. <https://english.cw.com.tw/article/article.action?id=3179>
4. <https://www.bundesbank.de/en/tasks/topics/economic-risks-from-germany-s-ties-with-china-922490>

Seminar on EU's Instantaneous Events

Social Security and Democracy

Second Reading-Report



Department of Computer Science and Information

Technology

Min-Sheng, Tsai-蔡旻盛

The European Union (EU) plays a crucial role in shaping social security policies and promoting democratic values among its member states. In the face of instantaneous events, such as crises, emergencies, or rapid societal changes, the EU's ability to uphold social security and democracy becomes paramount. Reflecting on this intersection prompts considerations about the EU's mechanisms for response, its commitment to democratic principles, and the challenges it faces in maintaining social cohesion and political stability during turbulent times.

One of the key strengths of the EU lies in its capacity to coordinate responses to instantaneous events that affect multiple member states. Whether it's a financial crisis, a public health emergency, or a humanitarian disaster, the EU can mobilize resources, expertise, and solidarity mechanisms to provide support and assistance to affected regions and populations. Through initiatives like the European Solidarity Corps and the European Social Fund, the EU promotes social cohesion and resilience by fostering cooperation and mutual aid among member states.

At the same time, the EU is committed to upholding democratic values and principles, including the rule of law, human rights, and transparent governance. In times of crisis, however, these values may come under pressure as governments grapple with urgent challenges and make decisions that impact civil liberties, political freedoms, and accountability mechanisms. Ensuring that emergency measures are proportionate, temporary, and subject to democratic oversight is essential to safeguarding democratic norms and institutions in the EU.

Moreover, instantaneous events can exacerbate existing social inequalities and vulnerabilities, posing risks to social security and political stability. Economic downturns, natural disasters, or geopolitical tensions can disproportionately affect marginalized communities, deepen Social divisions, and erode trust in democratic institutions. Addressing these challenges requires proactive policies that prioritize social inclusion, economic resilience, and democratic governance at both the national and EU levels.

From a personal perspective, the EU's response to instantaneous events prompts reflection on the interconnectedness of European societies and the importance of solidarity, cooperation, and democratic values in times of crisis. As citizens of the EU, we have a stake in ensuring that our collective response to emergencies upholds the principles of social justice, human dignity, and democratic accountability. Whether through civic engagement, advocacy, or community support, individuals can contribute to building a more resilient and inclusive Europe that prioritizes the well-being and rights of all its citizens.

In conclusion, the EU's approach to social security and democracy in the face of instantaneous events reflects its commitment to solidarity, resilience, and democratic values. By leveraging its collective resources and expertise, the EU can mitigate the impact of crises, uphold social cohesion, and strengthen democratic governance across member states. However, sustaining these efforts requires ongoing vigilance, engagement, and cooperation to address emerging challenges and build a more prosperous and democratic Europe for future generations.

Reference:

1. <https://docs.google.com/forms/d/e/1FAIpQLSfIzUINKGbGjwarL9K9vQAl6Q0PNBD1AkDTyomeMgjieZbijg/viewform>

Data Science and Deep Learning in Germany,
the UK, and Finland



Department of Computer Science and Information
Engineering
Min-Sheng, Tsai

Table of Contents

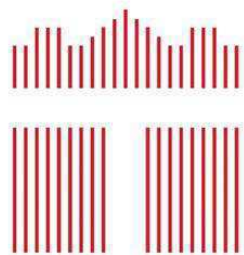
Introduction.....	3
Content.....	4
Germany: Pioneering Industrial Applications	4
Technical University of Munich (TUM)	4
Technische Universität Berlin (TU Berlin)	4
RWTH Aachen University	4
Merantix	5
United Kingdom: A Hub for AI Innovation	5
University of Cambridge	5
University of Oxford	6
Imperial College London.....	6
University College London (UCL).....	6
Meta AI (formerly Facebook AI Research).....	6
Google AI (DeepMind)	6
OpenAI	7
Finland: A Leader in AI Education and Implementation	7
Aalto University	8
University of Helsinki	8
Tampere University	8
Nokia's AI Initiatives.....	8
Wolt, The food delivery industry	8
Graph_Survey_In2023	9
Conclusion	12
Reference	13

Introduction

The European Union (EU) is actively pursuing AI development and integration into various sectors, including the economy. However, there isn't a specific "[AI generation](#)" in the Euro currency system. The Euro is the official currency of 19 out of the 27 EU member states, and its value is influenced by various economic factors, including but not limited to AI.

That said, AI technologies are increasingly being utilized in finance and economics across the Eurozone. For example, AI algorithms are employed in [financial trading](#), [risk assessment](#), [fraud detection](#), and [customer service](#) in banks and financial institutions operating within the Eurozone. Additionally, AI is being used to optimize [supply chains](#), improve productivity, and enhance decision-making processes in industries across the EU, which indirectly impacts the Euro economy.

EU institutions also have initiatives to promote AI development. [The European Commission has released strategies and investment plans to boost AI research, innovation, and deployment across member states.](#) These efforts aim to ensure that Europe remains competitive in the global AI landscape and can leverage AI technologies to drive economic growth and societal progress.



**National
Taiwan
University**
國立臺灣大學

Content

Data Science and Deep Learning are at the forefront of technological innovation, transforming industries and economies globally. Germany, the United Kingdom, and Finland stand out as pioneers in these fields, each with distinct approaches and significant contributions. This article explores the state of [Data Science](#) and [Deep Learning](#) in these three countries, highlighting key developments and their overall impact.

Germany: Pioneering Industrial Applications

Germany is renowned for its strong industrial base, where Data Science and Deep Learning are integral to driving innovation. The country is a leader in the Industry 4.0 revolution, leveraging these technologies to enhance manufacturing processes, predictive maintenance, and supply chain optimization. In the automotive sector, giants like [Volkswagen](#), [BMW](#), and [Daimler](#) are at the forefront, using AI for autonomous driving, vehicle diagnostics, and smart manufacturing.

Technical University of Munich (TUM)

TUM is renowned for its excellence in engineering and technology education. The university's Department of Informatics and affiliated institutes are leaders in AI research. [TUM collaborates closely with industry partners and participates in national and international AI projects, focusing on robotics, machine learning, and autonomous systems.](#)

Technical University of Berlin (TU Berlin)

TU Berlin is a prominent research university known for its contributions to AI and computer science. The university's Department of Electrical Engineering and Computer Science conducts cutting-edge research in AI, data analytics, and computational intelligence. [TU Berlin's interdisciplinary approach fosters innovation and industry collaboration in AI-driven technologies.](#)

RWTH Aachen University

RWTH Aachen University is a pioneer in engineering and technical education, offering comprehensive programs in computer science and AI. [The university's Chair of Computer Science 5 \(Information Systems\) focuses on AI research areas such as machine learning and intelligent systems.](#) RWTH Aachen's partnerships with industry and research institutes drive advancements in AI applications across various sectors.

Merantix

Merantix is a leading AI company based in Berlin, specializing in applied AI solutions for healthcare, autonomous systems, and industrial applications. The company collaborates with academic institutions and industry partners to develop AI-driven products that address complex challenges in sectors such as medical imaging, autonomous vehicles, and predictive maintenance.

Research institutions such as the Max Planck Institutes play a crucial role in advancing AI and Data Science. These prestigious institutions, alongside collaborations between universities and industries, drive applied research in these fields. The German government's AI strategy, which includes significant investments in AI research and development, aims to position Germany as a global leader in AI technologies.

Germany's AI Leadership

Germany's commitment to AI education and innovation, exemplified by institutions like TUM, TU Berlin, RWTH Aachen, and companies like Merantix, positions it as a global leader in AI technology. These entities drive advancements in AI research, industry adoption, and technological innovation, contributing to Germany's competitive edge in the global AI landscape.

United Kingdom: A Hub for AI Innovation

The United Kingdom is a global hub for AI and Data Science, driven by strong academic institutions, a vibrant startup ecosystem, and supportive government policies. Leading universities, including the University of Cambridge, University of Oxford, and Imperial College London, are renowned for their cutting-edge research in Data Science and Deep Learning. The Alan Turing Institute, the UK's national institute for data science and AI, plays a pivotal role in driving research and innovation across various sectors.

University of Cambridge

The University of Cambridge's Department of Computer Science and Technology is a leader in AI research, offering a range of AI-related courses. The university hosts several research groups focused on machine learning, natural language processing, and computer vision. Additionally, the Leverhulme Centre for

the Future of Intelligence (CFI) at Cambridge explores the implications of AI on society, aiming to ensure that AI development benefits humanity as a whole.

University of Oxford

Oxford University's Department of Computer Science boasts a strong AI program, with courses in machine learning, robotics, and data science. The Oxford Internet Institute conducts extensive research on the social and ethical implications of AI and technology. Oxford is also home to the Future of Humanity Institute, which investigates global challenges, including those posed by AI advancements.

Imperial College London

Imperial College London's Department of Computing offers comprehensive AI courses and engages in groundbreaking research in machine learning, robotics, and bioinformatics. The college's Data Science Institute serves as a hub for AI applications across various disciplines, fostering interdisciplinary research and collaboration. Imperial's commitment to AI research is reflected in its partnerships with industry leaders and its emphasis on real-world applications of AI technologies.

University College London (UCL)-Reference_link_has_teaching_course

UCL's Department of Computer Science is a significant contributor to AI research, with strengths in machine learning, computational statistics, and AI ethics. The UCL Centre for Artificial Intelligence is a leading research center dedicated to advancing AI technologies and their applications. UCL's AI research spans a wide range of fields, from healthcare to security, demonstrating its broad impact on society.

The UK is also home to several leading AI organizations that significantly contribute to the global AI landscape:

Meta AI (formerly Facebook AI Research)

Meta AI, previously known as Facebook AI Research (FAIR), has a strong presence in London. The London office focuses on advancing the state of the art in AI through open research and collaboration. Key areas of research include natural language processing, computer vision, and reinforcement learning. Meta AI aims to develop AI systems that can understand and interact with the world more naturally and effectively.

Google AI (DeepMind)

Google AI's London office is home to DeepMind, one of the world's leading AI research labs. Acquired by Google in 2015, DeepMind is renowned for its

breakthroughs in deep learning and reinforcement learning. Notable achievements include the development of AlphaGo, the first AI program to defeat a human professional Go player, and AlphaFold, which made significant advancements in protein folding predictions. DeepMind continues to work on a range of projects aimed at solving complex problems and improving AI's capabilities in various domains.

OpenAI

While OpenAI is primarily based in San Francisco, it has a growing presence in London. OpenAI focuses on ensuring that artificial general intelligence (AGI) benefits all of humanity. The London office supports research and development efforts in AI, contributing to OpenAI's mission to create safe and beneficial AI. OpenAI is known for developing advanced AI models, such as GPT-3, which powers this conversation.

London's status as a global AI hub is strengthened by leading AI organizations and a rich ecosystem of academic institutions, startups, and industry leaders. The UK's supportive regulatory environment and focus on ethical AI development make it an attractive destination for AI research. Notable startups like DeepMind thrive here, supported by a vibrant venture capital scene and government initiatives such as the AI Sector Deal, which underscore the UK's commitment to AI research, skills development, and ethical deployment.

Finland: A Leader in AI Education and Implementation

Finland is emerging as a leader in AI education and practical implementation, with a focus on accessible learning and innovative applications. The launch of the "Elements of AI" course, a free online course designed to educate citizens about AI, is a testament to Finland's commitment to AI literacy. This course has been translated into multiple languages and adopted by countries worldwide, significantly contributing to global AI education.

Finnish universities offer specialized programs in Data Science and AI, fostering a skilled workforce ready to tackle future challenges. In healthcare, Finland leverages Data Science and Deep Learning for predictive diagnostics, personalized medicine, and efficient healthcare delivery. AI is also used to optimize public services, including transportation, energy management, and urban planning.

Aalto University

Aalto University is renowned for its interdisciplinary approach to AI and Data Science. The university's Department of Computer Science offers a range of AI-related programs and is a key player in the Finnish Center for Artificial Intelligence (FCAI). Aalto's collaboration with industry and research centers fosters groundbreaking AI research in areas such as machine learning and computer vision.

University of Helsinki

The University of Helsinki is a pioneer in AI education and research. Its Department of Computer Science provides comprehensive AI programs and partners closely with the FCAI. The university's "Elements of AI" course, developed in collaboration with Reaktor, has had a global impact by making AI education accessible worldwide.

Tampere University

Tampere University excels in AI education and research, focusing on practical applications across various sectors. The university's Faculty of Information Technology and Communication Sciences offers specialized AI programs and engages in research projects that contribute to Finland's AI ecosystem.

Nokia's AI Initiatives

Nokia, a global leader in telecommunications, has been at the forefront of integrating AI into its operations. The company uses AI to optimize network performance, predict equipment maintenance needs, and enhance customer support services. Nokia's AI initiatives not only improve its business efficiency but also contribute to Finland's reputation as a hub for AI innovation.

Wolt: Leveraging AI and Data for Seamless Delivery

Wolt, The food delivery industry

Wolt, by leveraging AI and data science. a Helsinki-based technology company, Founded in 2014, Wolt uses advanced algorithms and machine learning to optimize delivery logistics, ensuring efficient and timely deliveries.

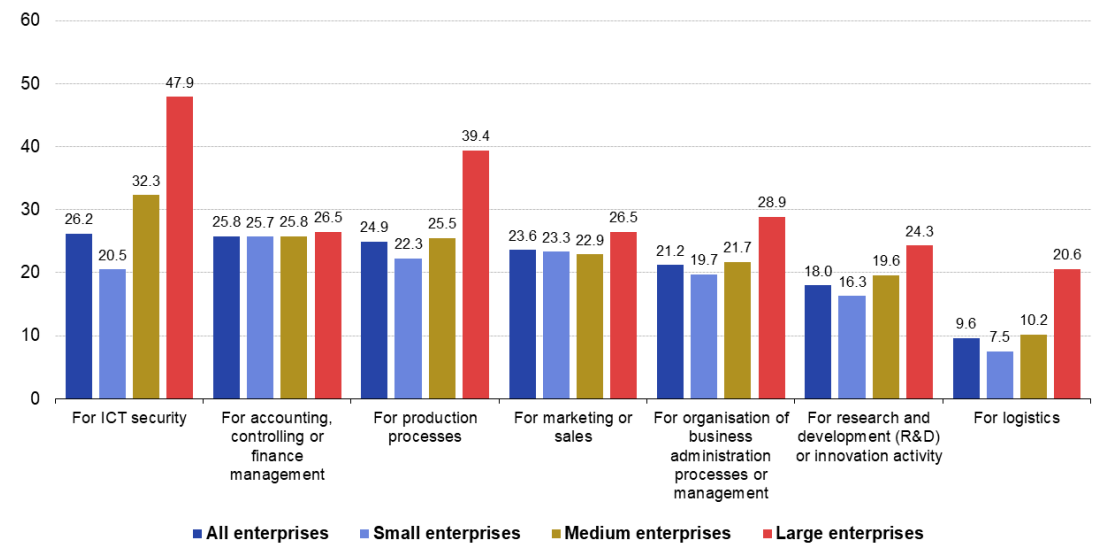
AI and Data-Driven Operations Real-Time Tracking and Predictive Analytics: Wolt's platform employs AI to provide real-time tracking of orders, allowing customers to follow their delivery from the restaurant to their doorstep. Predictive analytics help anticipate delivery times and manage logistics efficiently.

Finland's national AI strategy emphasizes collaboration between government, academia, and industry to drive AI adoption and innovation. Public-private partnerships are encouraged to accelerate AI implementation across various domains, ensuring that Finland remains at the forefront of AI advancements.

Graph_Survey_In2023

FIGURE 1

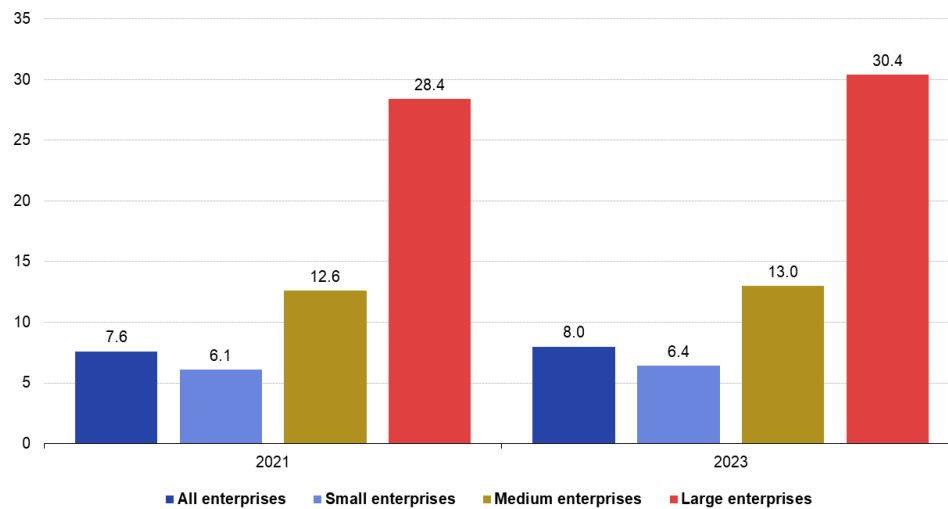
Enterprises using AI technologies by type of purpose and size class, EU, 2023
(% of enterprises using at least one AI technology)



Source: Eurostat (online data code: isoc_eb_ai)

FIGURE 2

Enterprises using AI technologies by size class, EU, 2021 and 2023
(% of enterprises)

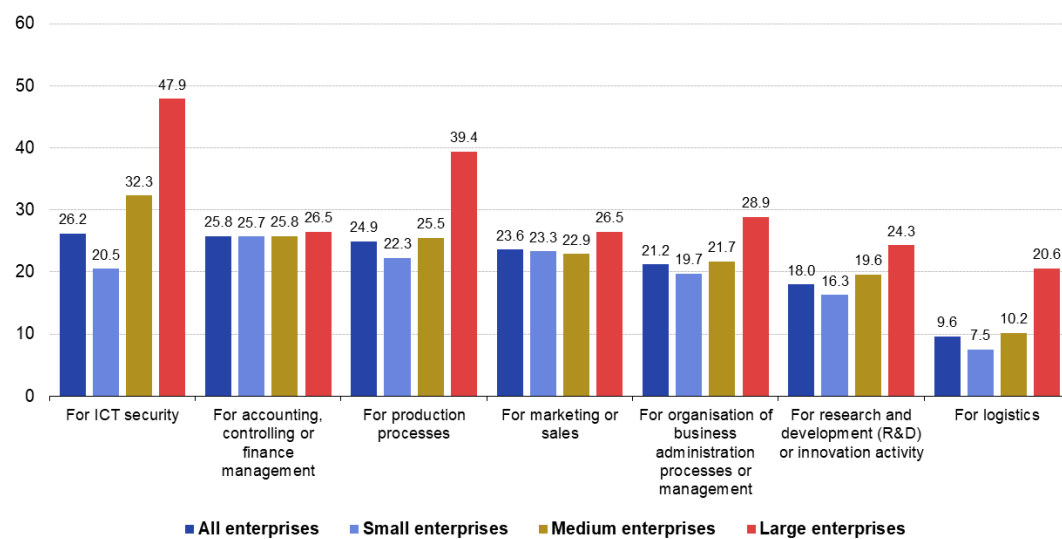


Source: Eurostat (online data code: isoc_eb_ai)

eurostat

FIGURE 3

Enterprises using AI technologies by type of purpose and size class, EU, 2023
(% of enterprises using at least one AI technology)



Source: Eurostat (online data code: isoc_eb_ai)

eurostat

FIGURE 4

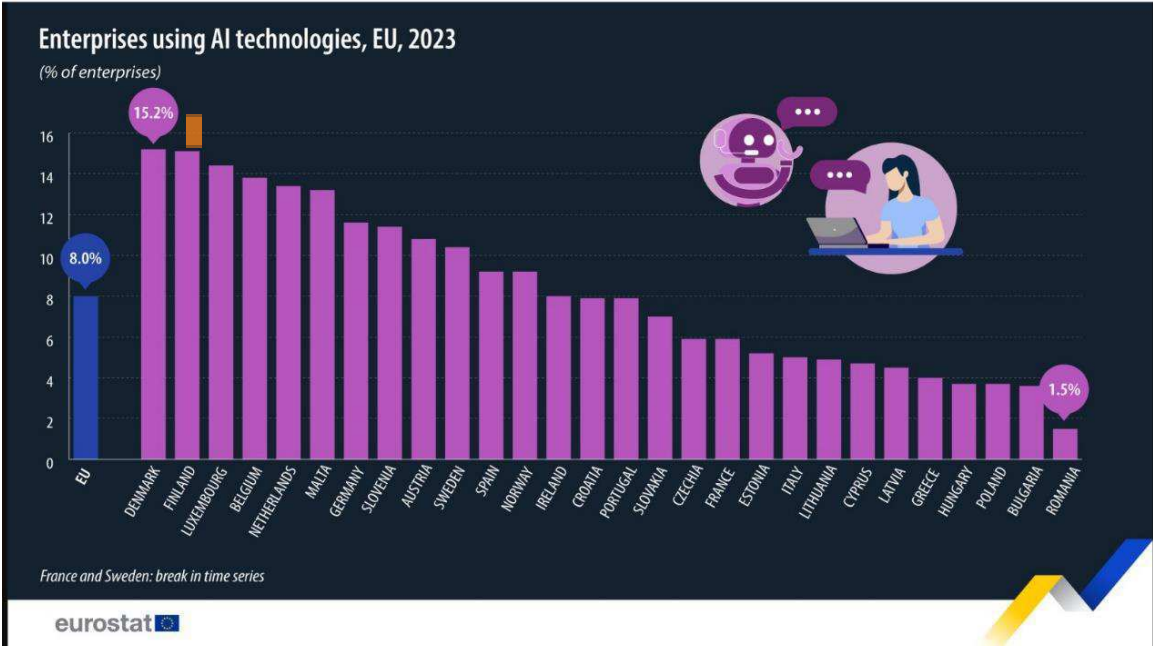
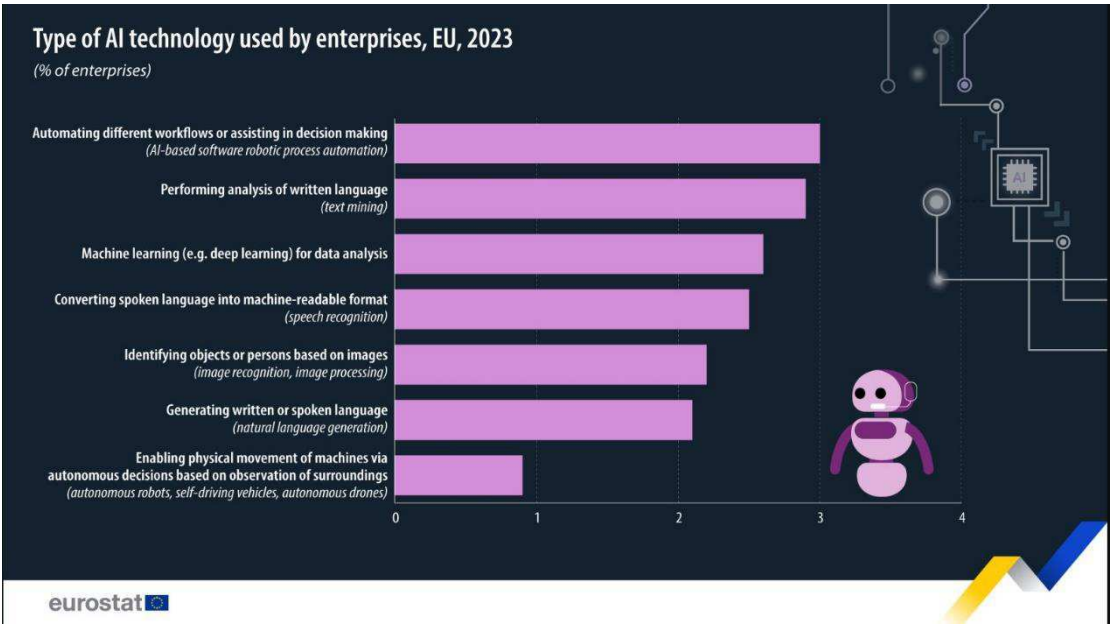


FIGURE 5



Conclusion

Germany, the United Kingdom, and Finland are at the forefront of advancements in Data Science and Deep Learning, each with unique strengths and strategic approaches. Germany's industrial prowess, the UK's innovation ecosystem, and Finland's focus on AI education and practical applications collectively contribute to the global AI landscape. These countries' efforts in research, industry collaboration, and government support ensure their continued leadership in Data Science and Deep Learning, driving technological progress and economic growth.

Reference

1. [Oxford-Artificial-Intelligence-Programme](#)
2. [Oxford-Ai-](#)
3. [Cambridge AI Labortory](#)
4. [Cambridge Acceleratescience](#)
5. [UCL AI-blog](#)
6. [UCL AI-center](#)
7. [University Of London-AI](#)
8. [University Of London-Queen Mary](#)
9. [TUM AI](#)
10. [TUM ai](#)
11. [TU Berlin AI](#)
12. [TU Berlin-Data Science](#)
13. [Aachen University AI center](#)
14. [RWTH-Aachen-University-Data-Protection](#)
15. [Aalto University FCAI](#)
16. [Aalto University Erasmus Mundus](#)
17. [University Of Helsinki AI](#)
18. [University Of Helsinki Data analysis](#)
19. [University Tampere AI hub](#)
20. [Finland Wolt](#)
21. [Finland NOKIA](#)
22. [Germany Openxcell](#)
23. [United Kingdom-Meta AI](#)
24. [United Kingdom Open AI](#)

25. United Kingdom Google AI